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Magnetostratigraphy of the Berriasian stratotype section (Berrias, France)

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The Berriasian (Early Cretaceous) stratotype section (Berrias, Ardèche, France), 30 m thick, consists of well-bedded blue-gray pelagic limestone. The biostratigraphic subdivision of this section is based on ammonite and calpionellid zonations. Magnetostratigraphic study has yielded a reliable magnetic polarity sequence. Magnetite is the principal ferrimagnetic mineral and the characteristic remanent magnetization (ChRM) directions are clearly defined after thermal or alternating field demagnetization. Ammonite and calpionellid zones allow biostratigraphic calibration of the magnetic polarity sequence with precision on the order of an ammonite sub-zone. This magnetic polarity sequence can be correlated with marine magnetic anomalies M15 to M17n. M15 occurs within the upper part of the *Boissieri* ammonite zone while M17n occurs within the *Occitanica* ammonite zone.

1. Introduction

Several attempts have been made to correlate magnetozones recorded in Jurassic-Cretaceous sedimentary sections with the M-sequence of marine magnetic anomalies. These magnetostratigraphic investigations were carried out on the Maiolica pelagic limestones in Italy [1–4], the Ammonitico Rosso limestone in the Southern Alps [5], the Sumeg section in Hungary [6] and deep-sea sediments (DSDP, Site 534) [7]. Age control of the above sections is based on calpionellid and/or nannofossil zonations, but the results obtained do not agree with each other. For example, Cirilli et al. [1] and Lowrie and Channell [3] correlated the Jurassic-Cretaceous boundary, recognized on the basis of calpionellids in Umbria Maiolica limestone, with the beginning of chron M17. But Ogg [5] recognized this boundary by the sharp increase in abundance of *Nannoconus colomii* in Lombardian Maiolica limestone and correlated it with the beginning of chron M18n. At DSDP Site 534, this event occurs at the top of the normal magnetozone correlated with chron M19n [7].

These apparent discrepancies, which were analyzed recently [8–10] result from inaccurate biostratigraphy and from different criteria for recognition of the Jurassic-Cretaceous (Tithonian-Berriasian) boundary. There are mainly two definitions of this boundary, (1) at the base of a combined *Grandis-Jacobi* ammonite zone which falls at the Calpionellid Zone A to B transition [11] and (2) between *Grandis* and *Jacobi* ammonite zones. This placement which coincides with the middle of Calpionellid Zone B is used in southeast France [12]. There are other definitions of the Jurassic-Cretaceous boundary using different biostratigraphic schemes such as nannofossil or dinoflagellate biozonations. These schemes are not well correlated with each other. Magnetostratigraphic studies can assist in the required correlations, and magnetostratigraphy of Early Cretaceous stratotype sections is of obvious importance to this problem.

This paper reports results obtained from the Berriasian stratotype section [13,14]. Because of

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the lack of faunas of the *Jacobi* ammonite zone, the Tithonian-Berriasian boundary is recognized in the Berriasian stratotype section as the middle of Calpionellid Zone B [15].

2. Lithostratigraphy, biostratigraphy, paleomagnetic sampling

The Berriasian stratotype section is located on the southeast margin of the Massif Central, 500 m north of the village of Berrias (44.38°N, 4.25°E) along Le Graveyron stream (Fig. 1). Oligocene tectonic disturbance of this region created tilt blocks, thus the Berriasian stratotype section is tilted 8° towards the southeast.

The Berriasian stratotype section is approximately 30 m thick, and consists of blue-gray micritic pelagic limestones. Beds of 10–50 cm thickness are sometimes separated by marly limestones 5–15 cm thick. In the lower part of the section, a 1.5 m thick breccia is overlain by cross-bedded limestone. The proportion of marly limestone increases towards the top of the section.

The biostratigraphic subdivisions of the Berriasian stratotype section are based on the work of Le Hegarat and Remane [15–17]. Their correlation of ammonite and calpionellid zones provides the biostratigraphic framework for this magnetostratigraphic study.

The Berriasian stratotype section was sampled at about 30–35 cm average spacing during the first collection (summer 1982), and a detailed resampling was undertaken during summer 1983. The site numbers refer to the sites of the first collection. The large gap between sites 20 and 25 results from poor outcrops along the stream. Because of poor outcrop, the Berriasian-Valanginian boundary has not been sampled.

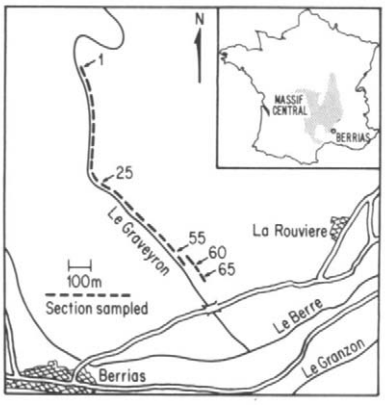


Fig. 1. Location map of the Berriasian stratotype section. Index map of France in upper right corner. Numbers in section refer to sampling sites.

3. Magnetic properties

3.1. Isothermal remanent magnetization

Coercivity spectrum analysis by acquisition of isothermal remanent magnetization (IRM) and subsequent thermal demagnetization is an effective technique for identifying the ferromagnetic content of pelagic limestones [18]. This method is necessary because the very low concentration of magnetic minerals makes extraction of adequate representative quantities for Curie temperature and other analytical methods difficult.

Ten samples were given IRM in increasing magnetizing fields up to 1.5 T with an electromagnet. The observed IRM acquisition of most samples suggests that the only carrier of the IRM is a low-coercivity (0–0.3 T) mineral, probably magnetite (Fig. 2a). Consistent with this interpretation, thermal demagnetization curves of the IRM reveal a maximum blocking temperature of 540°C.

A few samples located in the upper part of the section show a rapid initial increase up to 0.3 T due to magnetite followed by a slow increase up to 1.5 T (Fig. 2b). Thermal demagnetization shows a persistent component above 600°C. This high-coercivity and high blocking temperature component is probably hematite. However, the contribution of this mineral to the IRM is smaller.

Samples from the breccia in the lower part of the section present characteristics as shown in Fig. 2c. IRM acquisition reveals a high-coercivity component with a blocking temperature below 100°C. This high-coercivity mineral is most probably goethite. Above 100°C thermal demagnetization reduces the remaining IRM monotonously. The presence of an IRM component which saturates about 0.2 T suggests that a magnetite component may also be present.

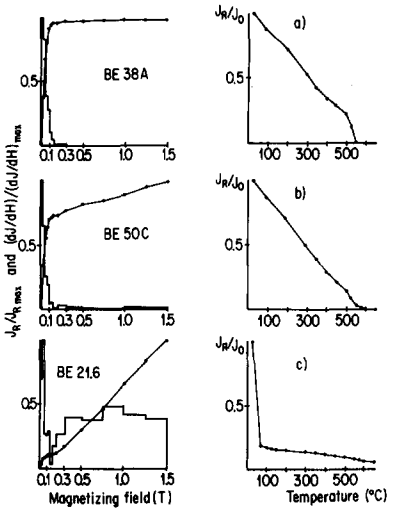


Fig. 2. IRM acquisition curves and thermal demagnetization of the maximum IRM. (a) IRM acquisition curves of most samples are characterized by saturation at fields of 0.2 T. The thermal demagnetization curves indicate a maximum blocking temperature of 550°C. This behavior indicates that magnetite is the only carrier of IRM. (b) In some samples, IRM increases rapidly up to 0.3 T. At higher magnetizing fields, a low gradient occurs up to 1.5 T. The thermal demagnetization indicates maximum unblocking temperature of 600°C. These components represent magnetite as the main carrier of IRM with a small contribution of hematite. (c) The rapid decrease of IRM during thermal demagnetization below 100°C shows that goethite is an important magnetic mineral in the breccia. In addition, the IRM acquired in magnetizing fields below 0.2 T and remaining above 100°C during thermal demagnetization is carried by magnetite. The final data points during thermal demagnetization were near the noise level of the magnetometer and are not significantly different from zero.

Since magnetite is the principal ferrimagnetic phase, it is probable that the natural remanent magnetization (NRM) is a detrital remanent magnetization (DRM) acquired at or soon after deposition.

3.2. *Measurement and analysis of the natural remanent magnetization*

The NRM measurements were carried out in the Laboratoire de Géomagnétisme (Université Paris VI) using a JR-4 spinner magnetometer and in the Laboratoire de Stratigraphie (Université Paris VI) using a 3-axis RS-01 (LETI/CEA) cryogenic magnetometer. Analysis showed that a magnetic intensity below 2.5×10^{-5} A/m could not be measured with good precision. The NRM intensities ranged from 5×10^{-5} to 9×10^{-3} A/m.

Following the initial collection, 62 specimens (1 specimen/site) were cleaned by progressive alternating field demagnetization. This technique removed an initial unstable component parallel to the present field in 20–25 mT peak fields, and isolated the characteristic remanent magnetization (ChRM) above 25 mT. The ChRM was parallel or antiparallel to the Early Cretaceous geomagnetic field at Berrias (Fig. 3a). Alternating field treatment was effective for 48 specimens. The remaining 14 specimens were either too weakly magnetized or anhysteretic remanence effects were observed above 20 mT.

165 specimens were cleaned by thermal demagnetization. Results obtained agree with those from alternating field treatment: an initial unstable component was removed by 300–350°C and, above this temperature, the ChRM vector was isolated (Fig. 3b). This treatment could not be used above 500°C because (1) most samples disintegrated and (2) a new magnetic phase was created above 400°C

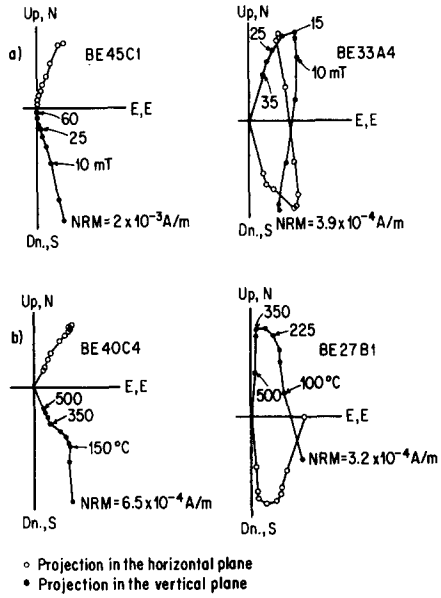


Fig. 3. Orthogonal projections [19] of (a) alternating field and (b) thermal demagnetization. An unstable component parallel to the present field is removed by 350°C or 25 mT. Above these values the ChRM is clearly defined.

perhaps due to alteration of pyrite which is ubiquitous in the section. Thermal treatment was effective for 115 specimens. Some specimens could not be adequately cleaned to isolate the ChRM because the intensity dropped below the noise level of the magnetometer.

No stable component could be isolated in specimens from the breccia in the lower part of the section: all measurable NRM was destroyed below 100°C, indicating that the natural remanent magnetization is due mainly to goethite.

Considering the magnetic properties of the samples, the unstable component is probably a viscous remanent magnetization (VRM) and the stable component is most probably a primary magnetization acquired by the sediment soon after deposition.

4. Magnetic stratigraphy

The ChRM directions are defined by averaging the remanence directions in the region of linear decay to the origin. These directions group in normal and reversed polarities (Fig. 4). These groups are not perfectly antiparallel suggesting that a small secondary component remains. The paleomagnetic pole position was calculated from the mean ChRM direction following structural correction (Table 1). For each sample, the latitude of the virtual geomagnetic pole (VGP) relative to the paleomagnetic pole was calculated from the ChRM direction. This relative latitude was used to determine the polarity of each sample (Fig. 5).

Polarity zones are clearly defined and provide a reliable magnetic polarity stratigraphy. This magnetic polarity sequence is biostratigraphically calibrated with ammonite and calpionellid zones. The Tithonian-Berriasian boundary could not be studied because the NRM intensity was too weak, thus the polarity sequence begins in the *Grandis* ammonite zone (Fig. 5). The Tithonian-Berriasian boundary is just 1.5 m below the base of the polarity sequence. A gap exists in the middle of the *Occitanica* ammonite zone, because no stable component could be isolated in the breccia and NRM intensity below this breccia was too weak. The polarity sequence ends in the *Callisto* am-

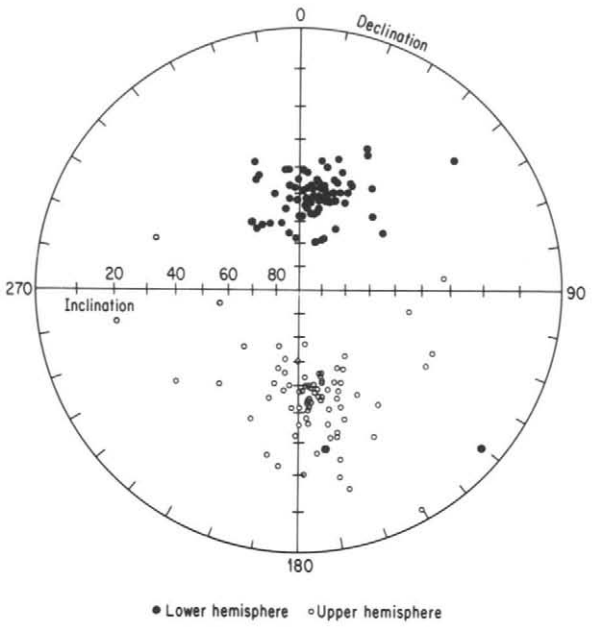


Fig. 4. Stereographic plots of the ChRM directions after magnetic cleaning and structural correction (163 specimens).

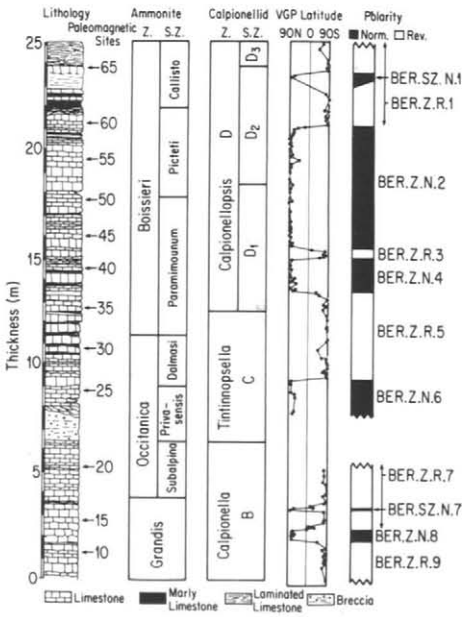


Fig. 5. Lithology, biostratigraphy and magnetic stratigraphy of the Berriasian stratotype section.

TABLE 1

Characteristic mean magnetization directions for the Berriasian stratotype section after structural correction

	N	D (°)	I (°)	α_{95} (°)	k
Normal	83	9.0	49.2	3.6	18.61
Reversed	80	170.6	−43.6	4.6	11.42
Total	163	0.4	47.3	2.9	15.75

Paleomagnetic pole calculated from sample mean directions: 74.1°N, 183.1°E; $dp = 3.1$, $dm = 4.2$
Paleomagnetic pole calculated from analysis of sample VGPs: 75.0°N 179.4°E; $\alpha_{95} = 3.2°$.

monite sub-zone. All polarity zones were labeled [20] using the following scheme:

BER: Berriasian
Z(SZ): zone or sub-zone
N(R): normal or reversed
1: number in the sequence

Two normal polarity zones were labeled as sub-zones (BER.SZ.N.1 and BER.SZ.N.7) because of their very short thickness, although BER.SZ.N.1 could be thicker, and few samples on which they are based.

5. Comparison with M-sequence of marine magnetic anomalies

Paleontologic results from basal sediments on oceanic crust above which the M-sequence of marine magnetic anomalies [21,22] is observed provide some constraint on the age of this oceanic crust. For example, DSDP Site 387 (western North Atlantic) was drilled on normal polarity oceanic crust between anomalies M15 and M16. The oldest sediments overlying basaltic basement at this site were found to be Berriasian or Early Valanginian by diverse paleontologic dating schemes: (1) *Cretarhabdus angustiforatus* nannofossil zone indicates Upper Berriasian–Early Valanginian [23], (2) *Biorbifera Johnewingii* dinoflagellate zone shows Upper Berriasian [24] and (3) Aptychi and Ammonites were found to be Berriasian [25]. Unless there is a large hiatus at the sediment-basalt interface, this constrains the age of chron M16n to be Berriasian. One of the purposes of this magnetostratigraphic study is to provide more detailed biostratigraphic calibration of this portion of the magnetic polarity time scale.

Accepting a Berriasian age for chron M16n, correlations between the polarity sequence in the Berrias section and magnetic polarity chrons M15 to M18 can be accomplished (Fig. 6 solid and dashed lines). BER.Z.R.1 is correlated with chron M15. A correlation between BER.SZ.N.1 and chron M15n seems improbable because of the short thickness of this polarity sub-zone. Magnetozones BER.Z.N.2 and BER.Z.N.4 represent chron M16n. Magnetozones BER.Z.R.5 and BER.Z.N.6 correspond to chron M16 and chron M17n. The boundary between the latter polarity interval and M17 could not be exactly determined because of the gap in the polarity sequence at Berrias. Below

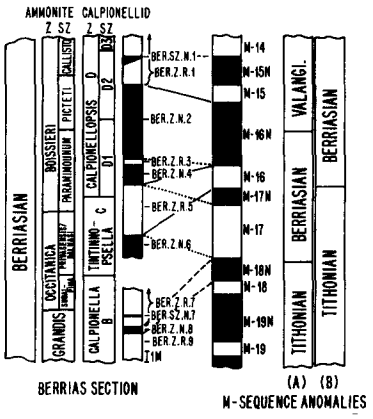


Fig. 6. Comparison of the magnetostratigraphy of the Berriasian stratotype section with the oceanic anomalies sequence M14 to M19 (A: Vogt and Einwich [22], B: Larson and Hilde [21]).

this gap, the correlations are somewhat ambiguous: BER.Z.N.8 could be correlated with chron M18n and thus BER.Z.R.9 could correspond to chron M18. Alternatively BER.Z.N.8 could be a previously unrecognized normal polarity subchron within chron M17, which like magnetozones BER.SZ.R.3 and BER.SZ.N.7, does not appear in the M-sequence. This is probably due to their short duration.

An alternative correlation scheme is represented by dotted lines (Fig. 6). BER.Z.R.3 is correlated with chron M16, BER.Z.N.4 with chron M17n, BER.Z.R.5 with chron M17 and magnetozone BER.Z.N.6 corresponds to chron M18n. This alternative scheme appears unlikely because (1) the short thickness of BER.Z.R.3 seems to exclude its correlation with chron M16 and (2) the basal portion of the section shows no reasonable correlation with the polarity pattern of chron M18 and/or M19.

6. Conclusions

The Berriasian stratotype section is the first Mesozoic stratotype on which a magnetic stratigraphy has been established. Magnetostratigraphic analysis of this section has yielded a reliable polarity sequence. The magnetic polarity sequence is calibrated by ammonite and calpionellid zones. In terms of ammonite zones, the polarity sequence begins in the *Grandis* zone and extends into the *Callisto* sub-zone of the *Boissieri* zone. The magnetozone corresponding to chron M15n occurs within the *Callisto* sub-zone of the *Boissieri* ammonite zone. The Upper portion of the magnetozone corresponding to chron M17n occurs within the *Privasensis* sub-zone of the *Occitanica* ammonite zone. Thus the biostratigraphic calibration of these magnetic polarity chrons has been refined to the ammonite sub-zone order.

The Jurassic-Cretaceous boundary could not be studied at Berrias, but its position is 1.5 m below the base of the polarity sequence. Depending on the correlation of the magnetostratigraphy with the M-sequence of polarity chrons, the Jurassic-Cretaceous boundary occurs either near the base of chron M18 if BER.Z.N.8 is correlated with chron M18n or near the base of chron M17 if BER.Z.N.8 represents a unrecognized normal polarity subchron within chron M17.

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